

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 32
- max_units: 128
- units_step: 32
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh', 'relu', 'elu']
- optimizers: ['adam', 'rmsprop', 'Nadam']

Tuning:

- max_epochs: 100
- hyperband_iterations: 3

LSTM Multi-Step Training Report

Report Generated: 2025-05-05 17:22:52

Training Start: 2025-05-05_16-01-14

Training End: 2025-05-05_16-08-27

Training Duration: 7.23 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.7310

MSE: 5.3473

RMSE: 2.3124

R2 Score: 0.0647

Multi-Step Forecast:

Forecast Steps: 20

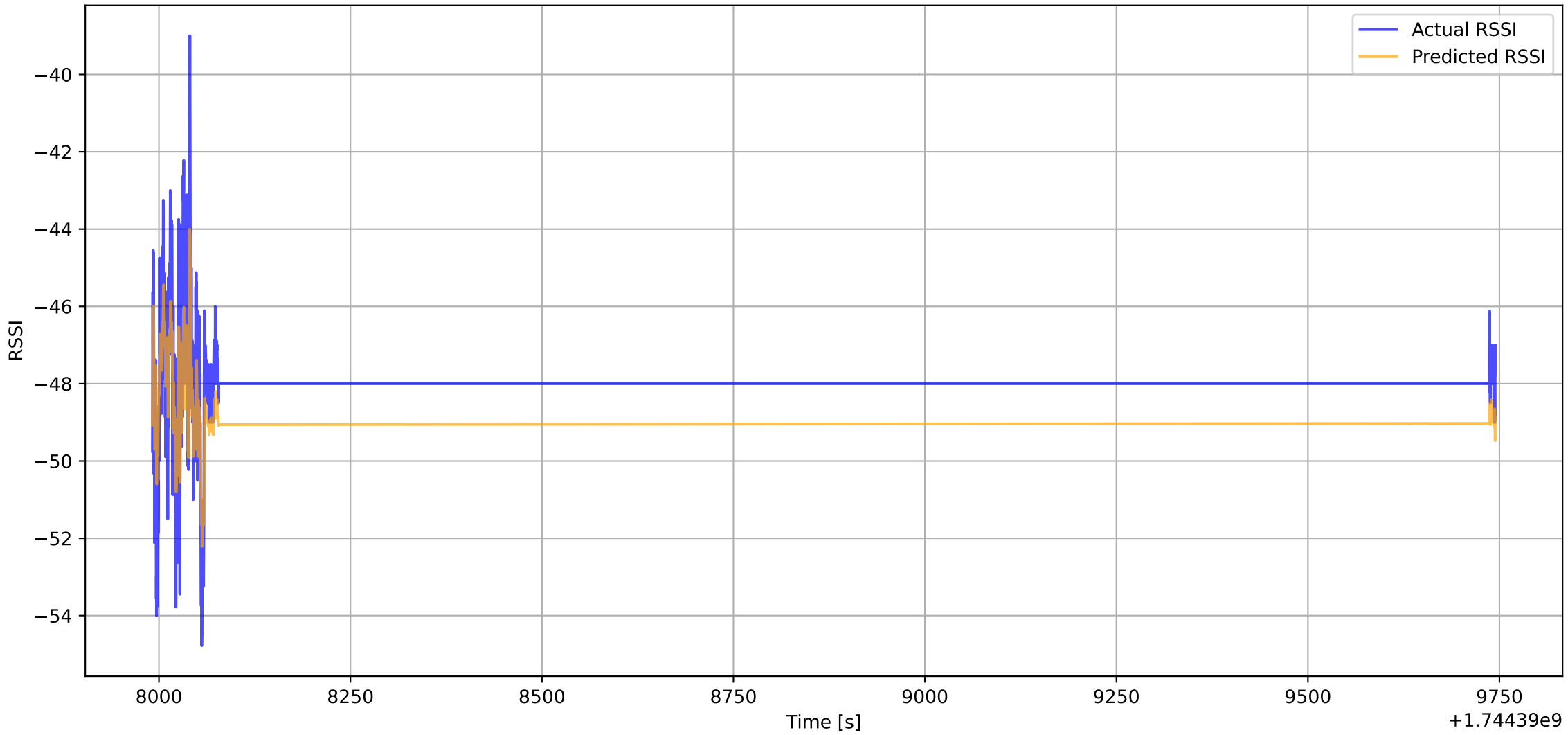
MAE: 0.0489

MSE: 0.0039

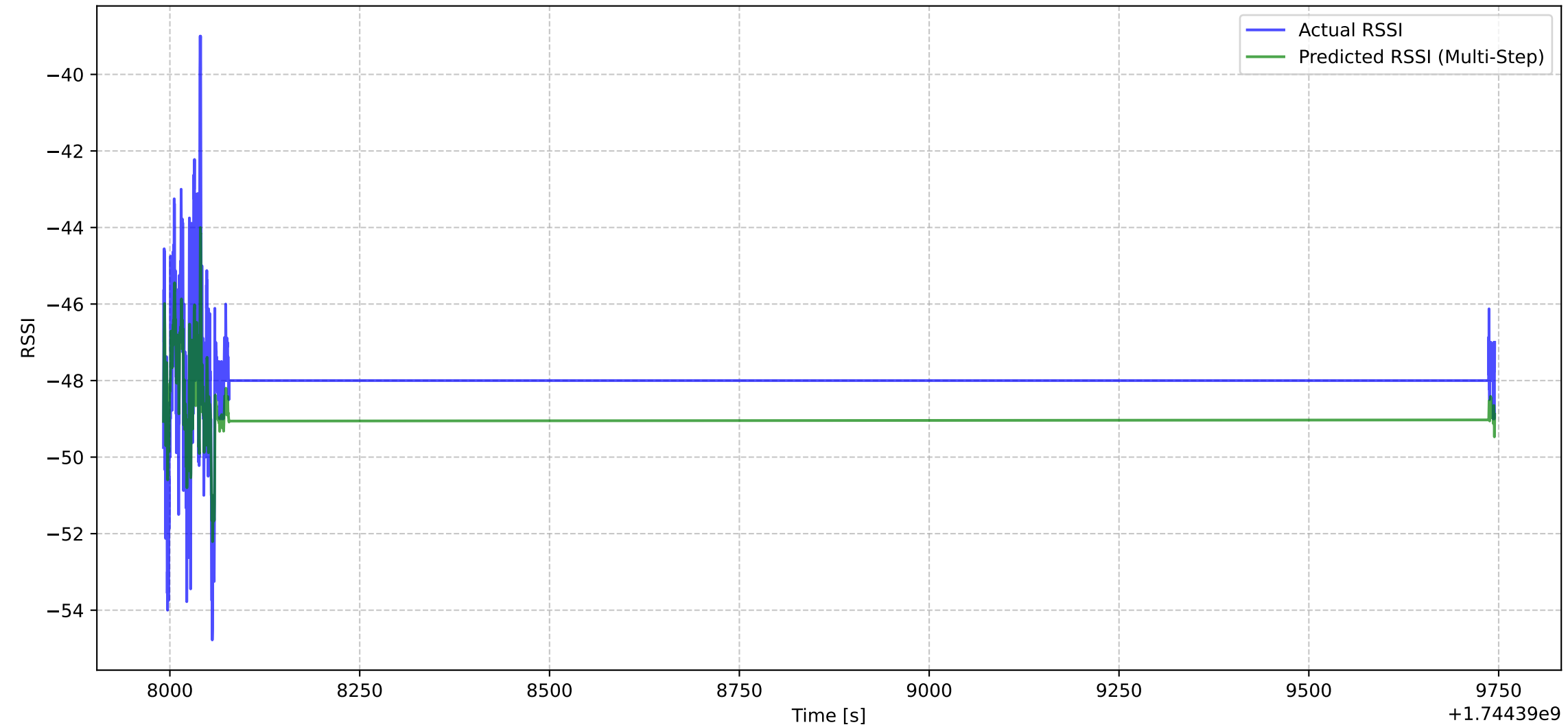
RMSE: 0.0622

R2 Score: 0.5192

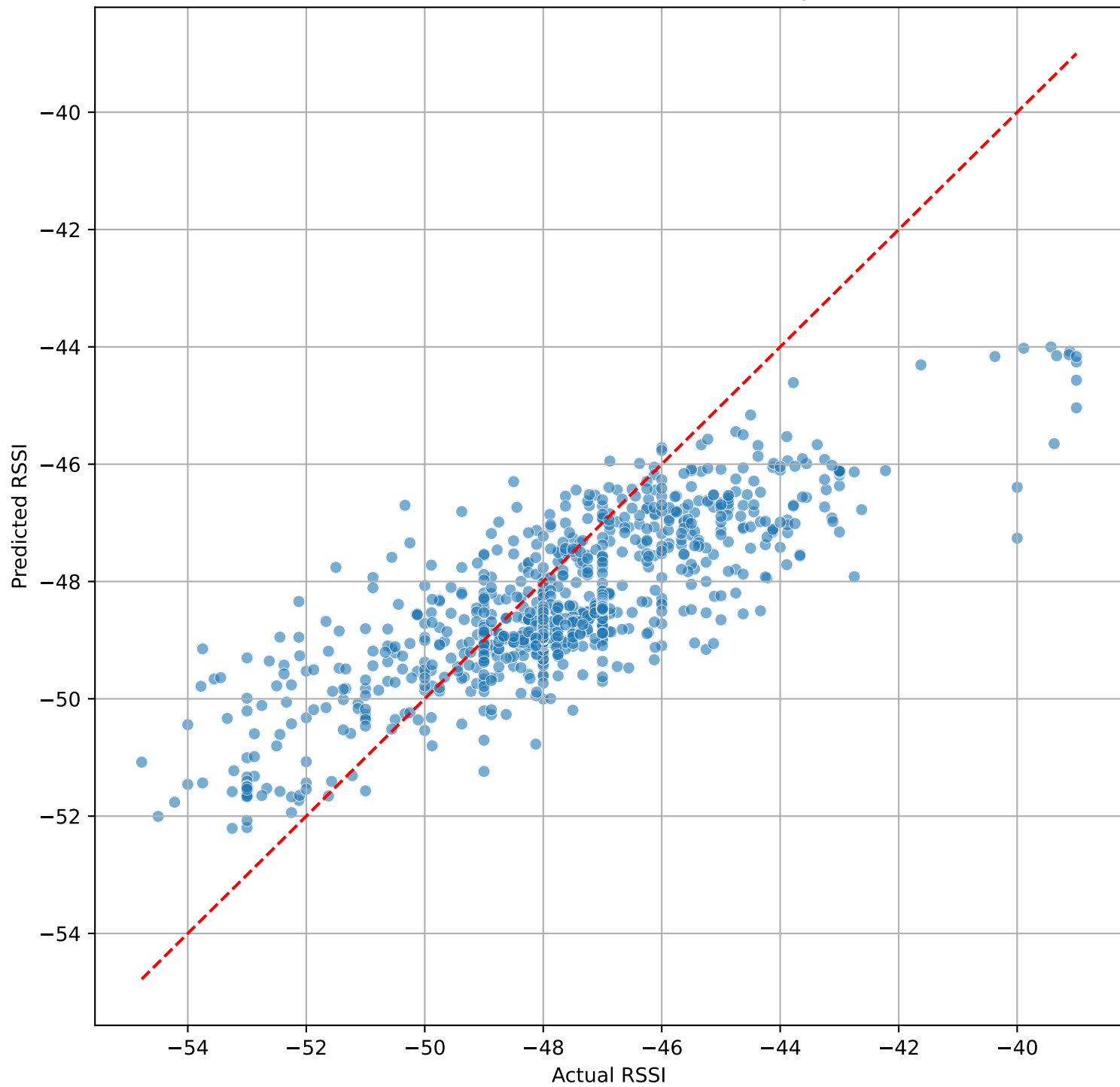
Actual vs Predicted RSSI (First Step)



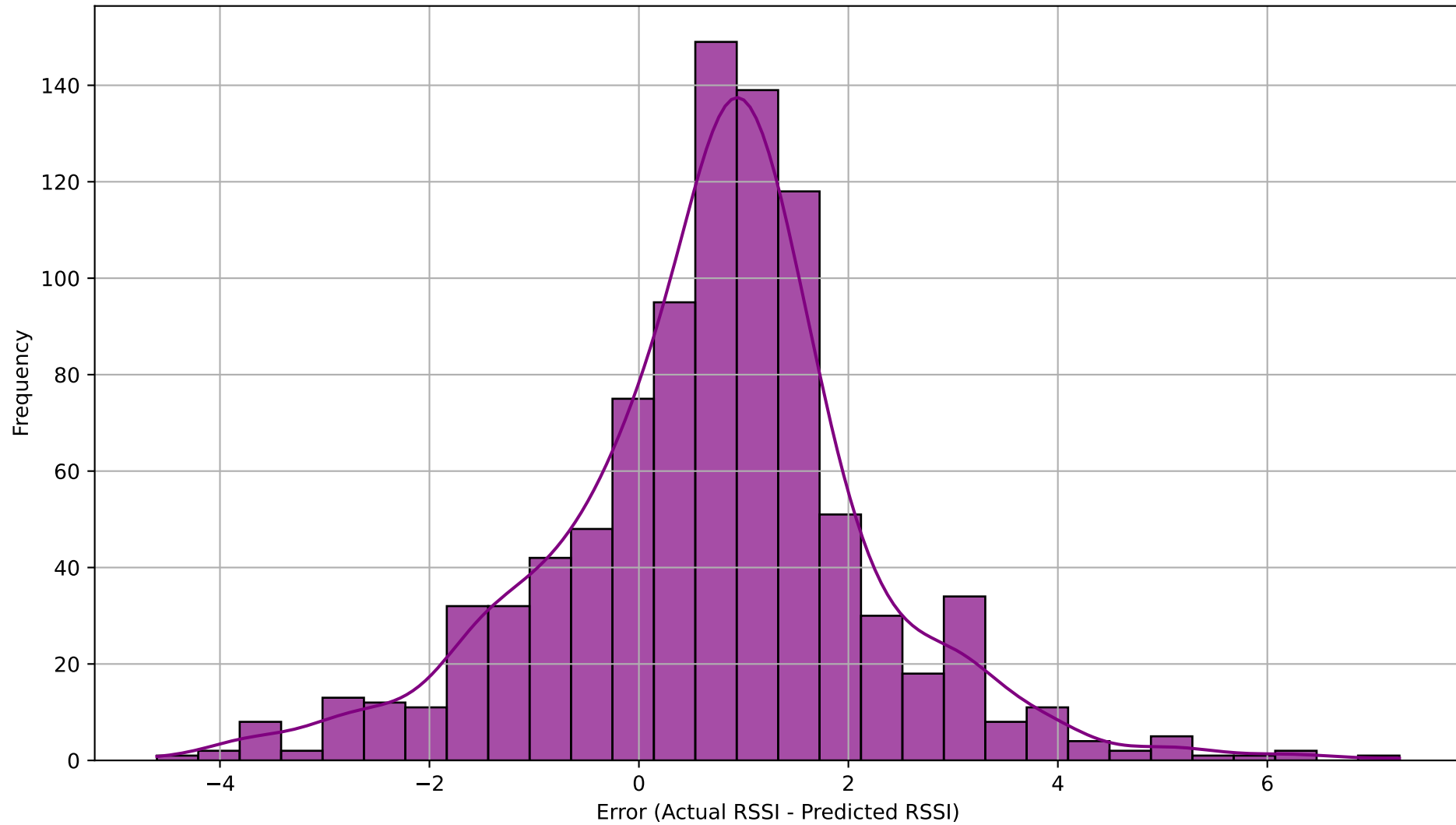
Actual vs Predicted RSSI (First Step vs Multi-Step)



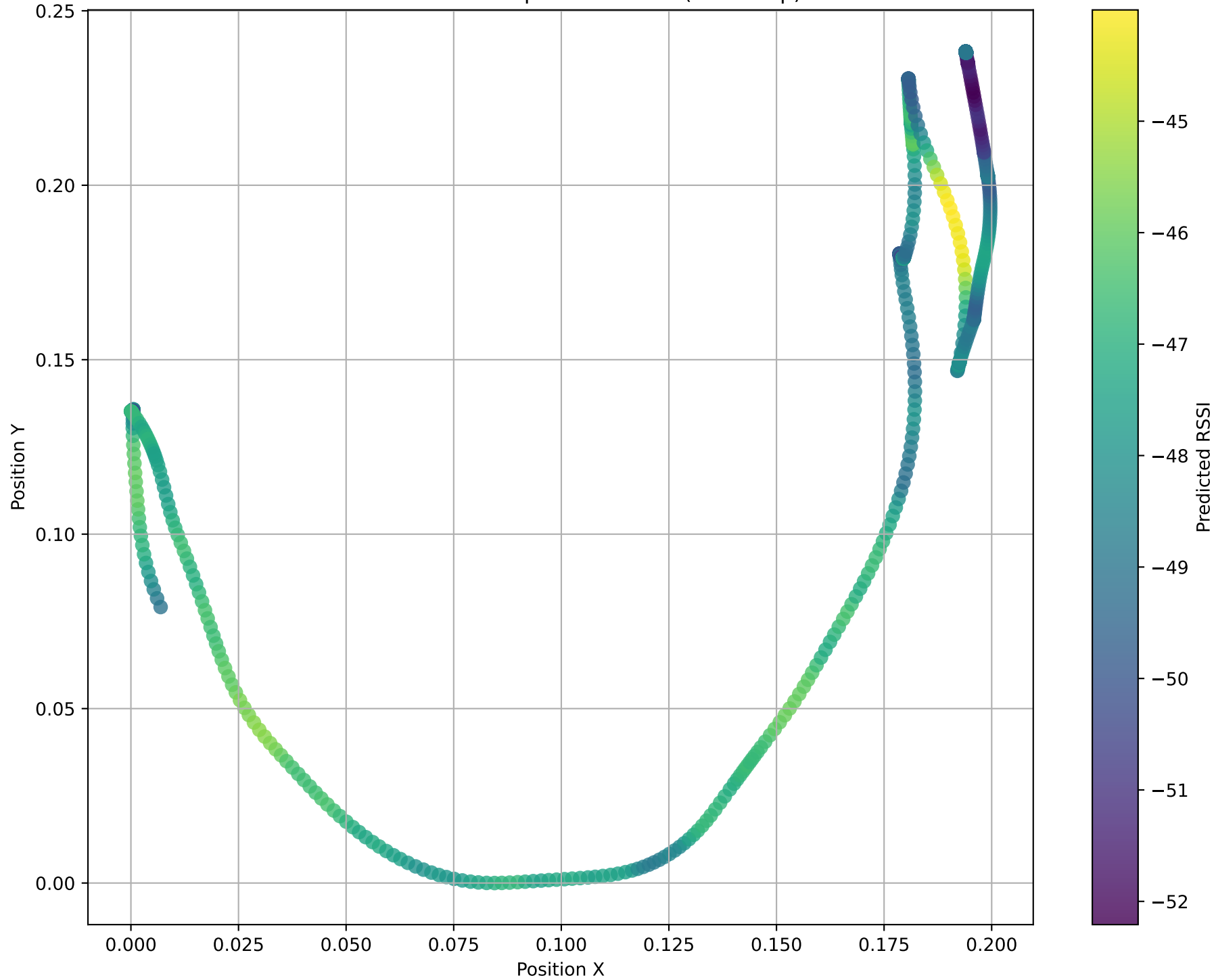
Actual vs Predicted RSSI (First Step)



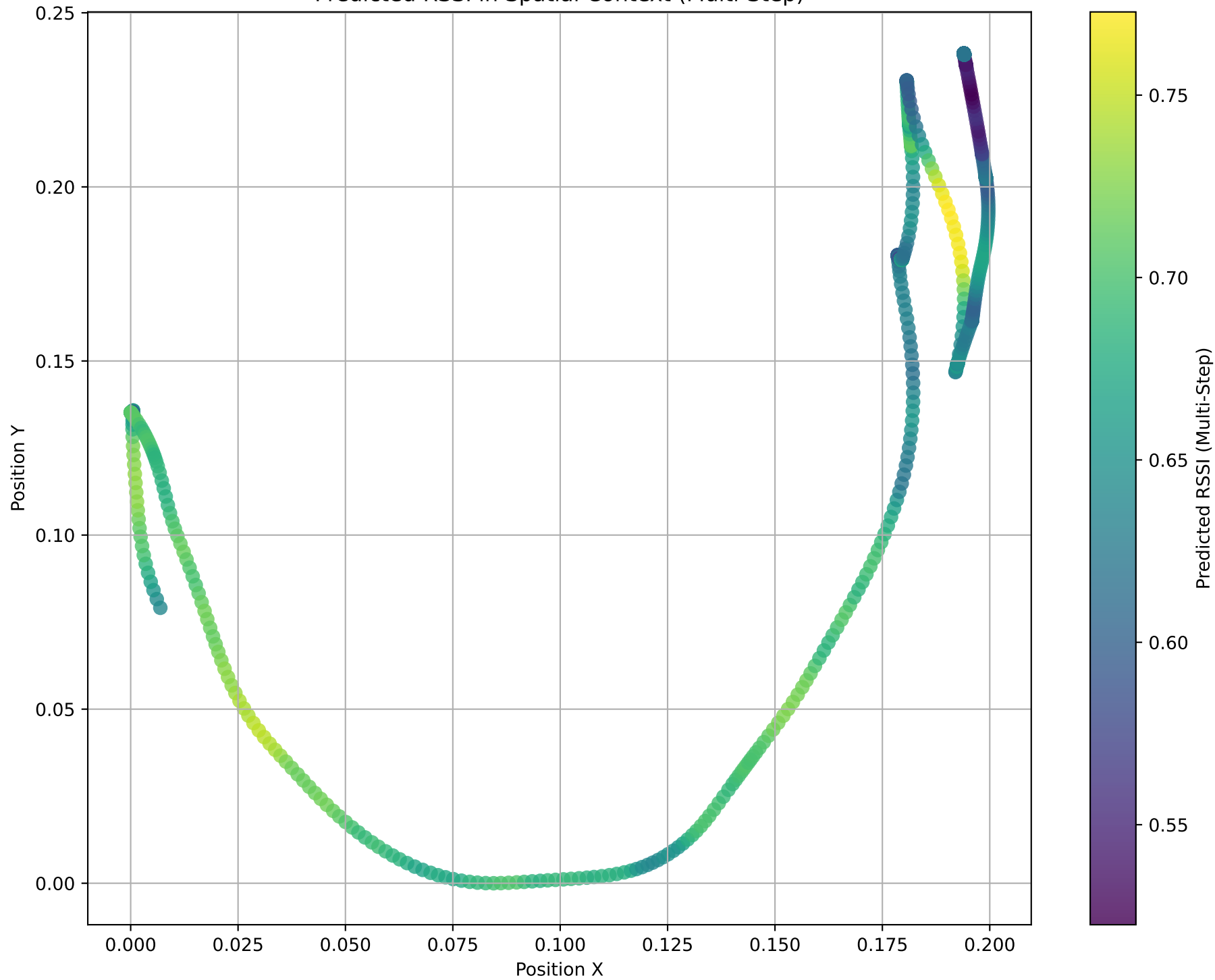
Distribution of Prediction Errors (First Step)



Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step

